

Waleed Ahmed

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Skills

Languages: C++, Python, Rust, Bash, SQL, TypeScript
Libraries & Frameworks: Robotics Middleware (ROS), Protobufs, Google Test, Pytest
Tools & Technologies: Git, Linux, Bazel, Profiling (Perfetto, Flamegraphs), Grafana, CI/CD, Claude Code, Cursor

Experience

Zoox

Foster City, CA

Software Engineer

Jun 2022 – Present

- Developing **C++** and **Python** infrastructure for integration test platforms running autonomy software on vehicle-representative hardware, using simulation and vehicle logs for validation, safety clearance, and latency testing; surfacing issues long before they reach the road
- Primary developer and maintainer of the playback framework behind a log-replay test that runs **thousands** of times a week to validate autonomy software changes; continuously modernized it onto the latest robotics middleware standards and expanded its capabilities to serve new use cases and teams across the org, while steadily improving its performance, reliability, and memory footprint
- Built and maintained **lidar, radar, and audio** sensor emulators that feed raw sensor packets to real drivers in the loop; expanded them for new sensors as the platform evolved, added fault injection for safety clearance, and eliminated a lidar fault affecting **60%** of runs
- Extended **robotics middleware** and orchestration frameworks to support offline testing, driving changes with framework owners to close gaps around diagnostics, resource limits, and configuration so vehicle software runs faithfully and reliably off-vehicle
- Software bring-up of test assets across **three compute generations**, expanding fleet capacity and cutting CI queue times by over **50%**
- Veteran on-call engineer and standing escalation point across teams; triaged issues, monitored test platform health, and uncovered a firewall gap silently blocking cross-region log replay, unlocking vehicle logs from **five additional cities**

Zoox

Foster City, CA

Software Engineering Intern

Sep 2021 – Dec 2021

- Developed software features for an integration testing framework capable of running autonomy software on representative hardware
- Designed and developed an improved sensor playback system that enabled more efficient and accurate playback during integration tests

Huawei

Toronto, ON

Autonomous Vehicles Software Engineering Intern

Sep 2020 – Dec 2020

- Developed a unit test suite for a Frenet frame motion planning stack with **90%** code coverage using **Google Test**
- Designed and implemented a path-building library using **C++** to construct a variety of reference paths to test planning algorithms on

Qualcomm

Toronto, ON

Automotive ADAS Software Engineering Intern

Jan 2020 – Apr 2020

- Developed system and application software for an ADAS and autonomous driving platform in **C/C++**
- Accelerated performance of a **computer vision SDK** by an average of **20x** by leveraging available hardware and software architecture in automotive focused Snapdragon SoCs

Christie Digital

Kitchener, ON

Software Engineering Intern

May 2019 – Aug 2019

- Developed user interfaces with **Qt** in **C++/QML** and bolstered code quality with setup of a **90%** code coverage **Jenkins** CI pipeline

Projects

Vizia

Capstone Project (Apr 2021-2022)

- Wearable glasses that can extract and communicate information from an image to a visually impaired user through audio transcription
- Consists of glasses with **Raspberry Pi** for image capture, **Flask** web app for computer vision processing, and an **iOS** app for audio playback

Autonomous Robot Racing (Software Lead)

UW Robotics (May 2018 - Aug 2019)

- Managed development for a robot that competed in the **International Autonomous Robot Racing Competition**
- Developed software architecture in **ROS** and **C++** for perception, mapping, and path planning using a stereo camera, IMU, and LiDAR

Education

University of Waterloo

B.ASc Computer Engineering

Sep 2017 – Apr 2022

- Notable Coursework: Distributed Computing, Programming for Performance, Robot Dynamics & Control, Autonomous Vehicles